

Project: - DAX Depo – Advanced Calculation Using DAX in Power BI

Objective

The main aim of this project is to understand sales performance and return behaviour using DAX in Power BI.

This helped in getting a better understanding of how DAX works with filter context, relationships.

Dataset Description

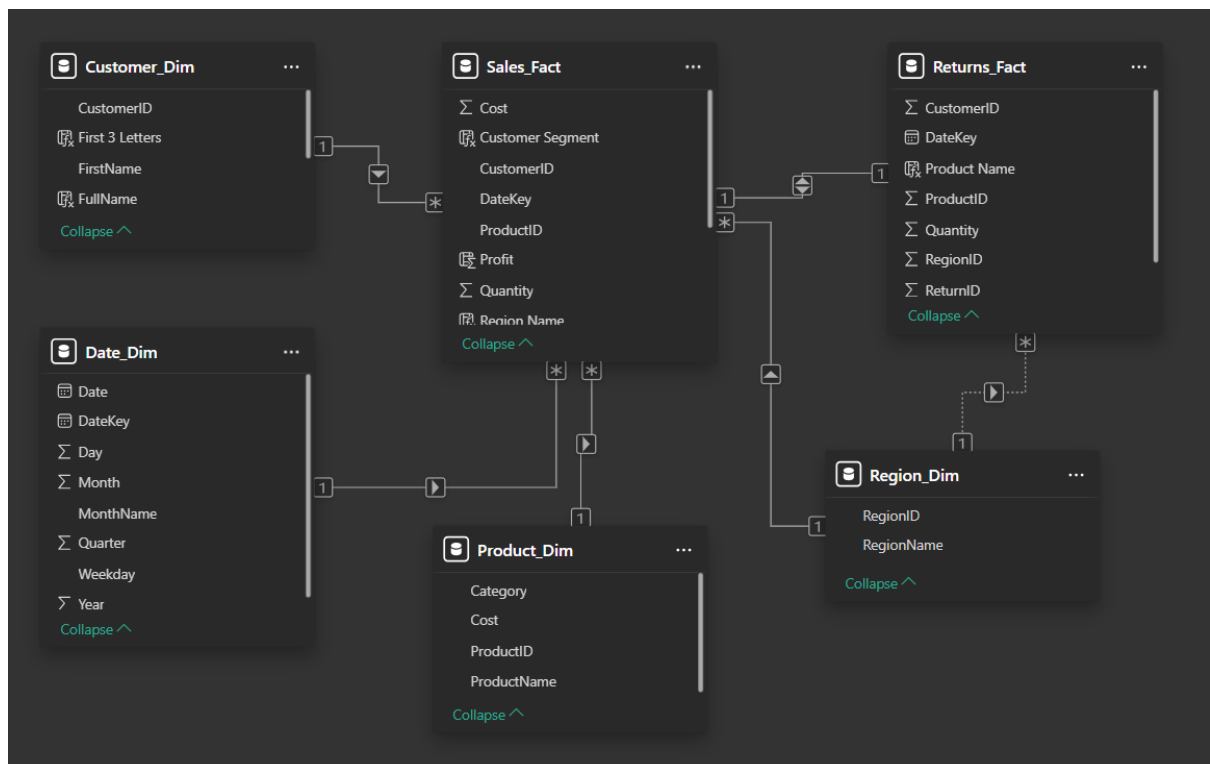
The dataset is divided into fact and dimension tables following a star schema design.

Fact Tables:

- **Sales_Fact** – contains all sales transactions including sales amount and cost
- **Returns_Fact** – contains records of returned orders

Dimension Tables:

- **Customer_Dim** – customer-related information
- **Product_Dim** – product details like category
- **Date_Dim** – used for time-based analysis (year, month, etc.)
- **Region_Dim** – geographical data



Calculated Columns

Some calculated columns were created to enrich the dataset:

- **Profit**
Calculated at row level by subtracting cost from sales.
- **Return Flag**
Used to identify whether a particular order was returned or not.
- **Customer Full Name**
Combined first name and last name for better readability in reports.

Profit	ReturnFlag	Sales Category	Region Name	Customer Segment
375	Not Returned	High	West	SMB
490	Not Returned	High	West	Corporate
135	Returned	High	West	Corporate
110	Not Returned	Medium	East	Retail
330	Not Returned	Medium	Central	SMB
145	Not Returned	Medium	South	SMB
155	Not Returned	High	North	Retail
40	Not Returned	Medium	North	SMB
200	Not Returned	High	East	SMB
275	Not Returned	Medium	South	Retail
190	Not Returned	Low	South	SMB
205	Not Returned	High	Central	Retail
385	Not Returned	High	South	Retail
225	Not Returned	High	East	SMB
255	Not Returned	High	Central	Retail
140	Not Returned	High	North	SMB
275	Not Returned	Medium	East	Corporate
300	Not Returned	High	South	Retail
95	Not Returned	High	North	Retail
460	Not Returned	High	Central	SMB
225	Not Returned	Medium	West	Retail
360	Returned	High	Central	SMB
490	Not Returned	High	South	SMB
190	Not Returned	High	East	Corporate
400	Not Returned	High	North	SMB
405	Not Returned	High	Central	Corporate
290	Not Returned	Medium	North	SMB
210	Not Returned	High	West	Corporate
80	Returned	High	North	Retail

FullName	Upper Name	First 3 Letters
Bethany Mcmillan	BETHANY	Bet
James White	JAMES	Jam
Karen Phillips	KAREN	Kar
Christopher Carter	CHRISTOPHER	Chr
Derrick Roman	DERRICK	Der
Samuel Simmons	SAMUEL	Sam
Jacob Crawford	JACOB	Jac
Jennifer Wheeler	JENNIFER	Jen
Daniel Lucas	DANIEL	Dan
Katherine King	KATHERINE	Kat
Kerry Smith	KERRY	Ker
Wendy Thompson	WENDY	Wen
David Bell	DAVID	Dav
Kathy Taylor	KATHY	Kat
Nicole Knight	NICOLE	Nic
Teresa Lewis	TERESA	Ter
Devin Morris	DEVIN	Dev
Jessica Sanchez	JESSICA	Jes
Lance Bailey	LANCE	Lan
Martha Jones	MARTHA	Mar
Amanda Thomas	AMANDA	Ama
Brad Calderon	BRAD	Bra
Johnathan Mathis	JOHNATHAN	Joh
Gary Woods	GARY	Gar

Measures

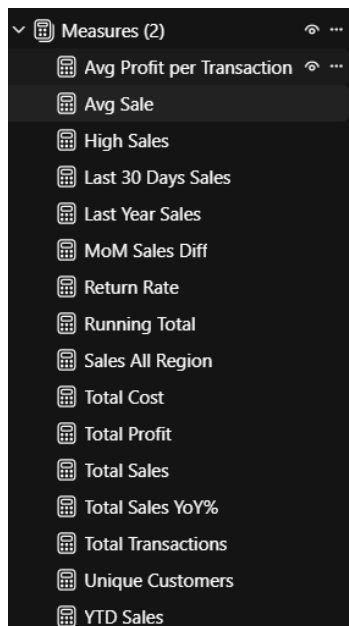
To analyse performance, several measures were created:

- **Total Sales** – sum of all sales
- **Total Cost** – total cost incurred
- **Total Profit** – difference between sales and cost
- **Return Rate**
Helps understand how many orders are being returned compared to total sales.
- **Average Sale per Transaction**
Gives an idea of how much revenue is generated per order.

Time-Based Calculations

Time intelligence functions were used to compare performance over time:

- **Year-over-Year Sales**
Used to compare current sales with last year's performance.
- **Month-over-Month Difference**
Shows how sales are changing month by month.
- **Year-to-Date (YTD) Sales**
Tracks cumulative sales within the year.
- **Running Total**
Helps in understanding overall growth trends over time.



Working with Filter Context

Some measures were created to understand how filters impact results:

- Sales ignoring filters (e.g., total across all regions)
- Sales for a specific region (like West)

This helped in understanding how CALCULATE, FILTER, and ALL functions behave.

Use of RELATED Function

The RELATED function was used to bring product category information into the sales table. This made it easier to use category-level analysis without creating complex joins in visuals.

Report Design (Matrix Only)

All outputs were displayed using Matrix visuals.

RegionName	Total Sales	Total Profit	YTD Sales	Return Rate	Running Total	Avg Sale	Last 30 Days Sales	FullName	First Segme	FullName	Product Name
Central	\$182,496	\$35,484	\$182,496	9.9%	\$182,496	\$899		Aaron Kelley	SMB	Aaron Kelley	Military
1	\$63,074	\$11,702	\$63,074	7.5%	\$63,074	\$941	63074	Adam Sullivan	Retail	Alan Wilkins	Ability
Books	\$7,827	\$880	\$7,827		\$7,827	\$978	7827	Alan Wilkins	Retail	Alicia Kelly	Debate
Clothing	\$3,300	\$819	\$3,300	33.3%	\$3,300	\$550	3300	Alexander Johnson	Retail	Amanda Thomas	Ability
Electronics	\$27,938	\$5,382	\$27,938	3.6%	\$27,938	\$998	27938	Alicia Kelly	Corporate	Andrew Nelson	Remember
Home Appliances	\$22,242	\$3,814	\$22,242	10.5%	\$22,242	\$1,171	22242	Amanda Thomas	Retail	Arthur Nichols	Debate
Toys	\$1,767	\$807	\$1,767		\$1,767	\$295	1767	Andrew Nelson	Retail	Becky Campbell	Debate
3	\$46,991	\$10,207	\$46,991	11.1%	\$46,991	\$746	46991	Anita West	Corporate	Bethany Mcmillan	Read
Books	\$13,470	\$1,861	\$13,470	40.0%	\$13,470	\$1,347	13470	Anthony McCarthy	Retail	Bonnie McLaughlin	Interest
Clothing	\$5,053	\$949	\$5,053		\$5,053	\$561	5053	Arthur Nichols	SMB	Brad Calderon	Miss
Electronics	\$19,378	\$4,128	\$19,378	4.8%	\$19,378	\$923	19378	Barbara May	Retail	Bryan Mckee	Media
Home Appliances	\$6,758	\$2,275	\$6,758	6.7%	\$6,758	\$451	6758	Becky Campbell	Retail	Catherine Bridges	Miss
Toys	\$2,332	\$994	\$2,332	12.5%	\$2,332	\$292	2332	Bethany Mcmillan	Retail	Charles Maxwell	Ability
11	\$72,431	\$13,575	\$72,431	11.0%	\$72,431	\$992	72431	Bonnie McLaughlin	Corporate	Christopher Carter	Short
Books	\$13,112	\$1,708	\$13,112	11.1%	\$13,112	\$1,457	13112	Brad Calderon	Corporate	Christopher Thompson	Miss
Clothing	\$4,376	\$1,100	\$4,376		\$4,376	\$729	4376	Brian Lopez	SMB	Claudia Johnson	Read
Electronics	\$25,291	\$2,706	\$25,291	10.5%	\$25,291	\$1,331	25291	Bryan Mckee	Retail	Daniel Lucas	Miss
Home Appliances	\$24,424	\$5,055	\$24,424	8.0%	\$24,424	\$977	24424	Carolyn Howe	Corporate	Danielle Moon	Want
Toys	\$5,228	\$3,006	\$5,228	21.4%	\$5,228	\$373	5228	Catherine Bridges	Retail	David Bell	Organization
East	\$151,499	\$29,568	\$151,499	11.1%	\$151,499	\$765		Charles Maxwell	Retail	David Browning	Ability
1	\$44,490	\$10,013	\$44,490	12.5%	\$44,490	\$695	44490	Christine Spence	SMB	Deborah Hood	Want
Books	\$10,512	\$1,754	\$10,512	8.3%	\$10,512	\$876	10512	Christopher Barron	SMB	Debra Johnson	According
Clothing	\$8,111	\$842	\$8,111	11.1%	\$8,111	\$901	8111			Derrick Roman	Organization
Electronics	\$13,732	\$3,789	\$13,732	9.5%	\$13,732	\$654	13732		Corporate	Destiny Bell	Media
Home Appliances	\$7,298	\$1,091	\$7,298	10.0%	\$7,298	\$730	7298			Devin Morris	Read
Toys	\$4,837	\$2,537	\$4,837	25.0%	\$4,837	\$403	4837			Edward Diaz	Media
3	\$56,326	\$11,357	\$56,326	14.1%	\$56,326	\$793	56326			Edward Shah	Anyone
Books	\$12,534	\$1,977	\$12,534	16.7%	\$12,534	\$1,045	12534			Francisco Mckenzie	Debate
Clothing	\$7,248	\$1,521	\$7,248		\$7,248	\$906	7248			Frank Baker	Happy
Electronics	\$18,770	\$3,008	\$18,770	14.3%	\$18,770	\$894	18770			Gary Woods	Read
Total	\$848,397	\$162,172	\$848,397	10.0%	\$848,397	\$848				Jacob Crawford	Want
										Total	Ability
								MonthName	SalesAmount	Profit	
								January	281569	55813	
								March	280086	53391	
								November	286742	52968	
								Total	848397	162172	